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Enhancing diagnosis: ensemble deep-learning model for fracture detection using X-ray images

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AIM: Orthopedic trauma results in the injury of bone joints and tendons of the body. A radiologist reviews and monitors large numbers of radiographs daily, which can lead to the diagnostic error. Therefore, there is a need to automate the detection of bone fractures in X-ray images, particularly humerus bone fractures. In this paper, we have proposed an ensemble model that can detect the fracture in an x-ray image.

MATERIALS AND METHODS: In this paper, we proposed an ensemble model designed for fracture detection in X-ray images. An ensemble model combines multiple diverse models to improve predictive accuracy and robustness by aggregating their individual predictions. The model leverages MobileNetV2, Vgg16, InceptionV3, and ResNet50, using histogram equalization for preprocessing and a Global Average Pooling layer for feature extraction. The entire humerus from the public Mura-v1.1 dataset is utilized for analysis, utilizing a single training-validation split. The dataset is divided into a ratio of 80:20 for experiments for the training and validation datasets.

RESULTS: The proposed model outperformed the modified deep-learning models and achieved 92.96%, 91.62%, and 92.14% accuracy, recall, and F1 scores, respectively.

CONCLUSION: The ensemble model presented effectively automates bone fracture detection in X-ray images of the humerus, demonstrating superior performance compared to modified deep-learning models. A comparison has been made between a novel ensemble model and state-of-the-art models, bench-marking their performance. These findings underscore its potential for enhancing diagnostic accuracy and efficiency in orthopedic radiology.

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Introduction

Bone fractures are a prevalent medical problem that imposes significant costs and requires resource utilization for treatment and monitoring. While a high proportion of fractures result from great impact or tension, they can also be associated with bone weakening caused by conditions such as osteoporosis or bone diseases such as cancer.[1] Substantial investments in X-ray technology and medical personnel aim to enhance patient care, yet current fracture diagnosis remains manual.[2] Modern digital radiography provides rich data underutilized in diagnostics, prompting the application of computer vision techniques.[3] These can automate fracture identification and categorization using machine-learning and deep-learning (DL) algorithms, leveraging image patterns for enhanced diagnostic efficiency.[4] Computer vision techniques are designed to support rather than replace healthcare professionals in diagnostics, aiding informed decision-making.[5] X-ray exposure risks are generally lower than procedures such as computed tomography (CT) or fluoroscopy.[6]

Analyzing X-ray images to identify bone fractures can pose challenges, when prompt access, to professionals is not readily available.[7,8] This can lead to overlooked diagnoses and delayed treatment, which can have an impact on outcomes. Errors in diagnosis in situations underscore the importance of precise and timely detection of fractures. Artificial intelligence (AI) learning technology presents a solution by automating the detection of fractures, with efficiency that matches or even exceeds human capabilities.[9,10] This helps enhance precision minimize errors and elevate the quality of patient care. Deep-learning models have produced positive outcomes in automating the identification and categorization of fractures from medical images, such as CT scans, X-rays, and magnetic resonance imaging (MRI) scans[11,12,13,14]. Fracture detection using deep learning is the focus of current research in medical imaging and computer vision. In this study, an innovative DL technique is proposed and used to identify fractures in the X-rays of humerus.

Related work

To create a technique for the categorization of fresh studies as either "fracture" or "no fracture," the Inception v3 network's top layer is retrained utilizing radiographs of the lateral wrist.[15] In paper,[16] dilated convolutions were used to create a new backbone network. A new DL approach, known as the dilated convolutional feature pyramid network (DCFPN), is devised and used to identify thigh fractures by swapping out the backbone network of the contemporary-feature pyramid network framework with the newly built one. A novel classification network, the two-stage fracture-sensitive CNN for Bone Fracture Detection by Yangling Ma, is applied.[17] Dimililer classified whether a bone is broken on X-ray. After the preprocessing stage, the neural network of the system is assembled.[18] Hagiwara *et al.* proposed a DL framework for bone age assessment using radiographs. The model used a convolutional neural network (CNN), which has great accuracy in predicting bone age and may be used to detect fractures among younger patients.[19] Wang *et al.* proposed the FracNet DL model for automated fracture diagnosis in X-ray images.[20] To increase localization accuracy, FracNet merged a CNN with a fully linked Conditional Random Field. The model successfully identified the fractures with good sensitivity and specificity. Zhang *et al.* created a network to automatically detect and categorize Wrist fractures in X ray images. By using a region-based CNN method, the model effectively recognizes fractures with precision, leading to expedited diagnosis and treatment strategies. Additionally, Cheng *et al.* explored the identification and classification of fractures in radiographs.[21] To achieve precise fracture localization and classification, researchers have devised a cascaded network design that integrates detection and classification networks. FractureNet is a DL framework for prevalent fractures. In Gertych *et al.*, the authors developed the FractureNet DL system for generic fracture diagnosis in X-ray images.[22] By combining a CNN architecture with attention mechanisms, FractureNet excelled at numerous datasets for fracture identification.

Computer-aided diagnosis using artificial intelligence (AI) can enhance healthcare professionals' capabilities and contribute to more effective patient care.[23,24] Deep learning is a field within machine learning that utilizes neural networks as its primary model architecture.[25] In DL, neural networks, particularly those with convolutions in CNNs, show great promise for analyzing medical images. What distinguishes DL from further machine-learning approaches is that it can detect by itself and extract optimal direct features from unfiltered raw data without requiring manual feature engineering.[26] To achieve optimal performance, DL models require a substantial amount of training data, a well-designed data preprocessing pipeline, and careful hyperparameter tuning.[27] An effective DL architecture for computer vision and image recognition tasks is a CNN.[28,29] The author developed an ensemble AI method for detecting hip fractures in AP pelvis plain radiographs, utilizing majority voting among three DL models (Xception, EfficientNet, and NfNet) to achieve the most reliable and precise diagnoses.[30] As CNNs and their derivatives have been effectively used in other disciplines, including processing natural language and speech recognition, their influence extends beyond computer vision.[31] The authors evaluated DenseNet-169, DenseNet-201, and InceptionResNetV2 models on the MURA dataset for detecting abnormalities in humerus and finger radiographs. DenseNet-201 and InceptionResNetV2 achieved high accuracies and sensitivities for humerus radiographs, but performance on finger radiographs was less promising due to inter-radiologist variation.[32]

The proposed technique

The goal of this research is to develop an effective methodology for classifying bone X-rays into two categories, fracture and nonfracture, to improve the accuracy. To achieve this objective, a novel ensemble model is introduced.

Novel ensemble model

A novel ensemble model has been proposed to enhance overall classification performance by integrating multiple CNN architectures, specifically MobileNetV2, InceptionV3, VGG16, and ResNet50. These architectures, pretrained on the extensive ImageNet dataset, are further refined by adjusting their model parameters and modifying their layers. Fine-tuning hyperparameters and adjusting network structures optimize model performance, including adding layers or freezing specific ones. The ensemble model integrates diverse strengths to enhance feature capture and pattern recognition, aiming for superior classification accuracy compared to individual model.

A block diagram of the proposed ensemble model is presented in Fig 1.

A stacked generalizer, also known as a meta-model in ensemble learning, combines predictions from CNN models such as MobileNetV2, InceptionV3, VGG16, and ResNet50. It uses a logistic regression or small neural network to refine and blend these predictions, leveraging their complementary strengths to improve classification performance. It takes the probability scores (P1, P2, P3, P4) represented as predictions from the base models as input features. In Fig 1, The stacked generalizer is trained on the validation set's probability scores to learn the optimal combination of these scores. During inference, the meta-model integrates outputs from base models to classify X-ray images from the Mura-v1.1 dataset as fractured or nonfractured. Each base model contributes probability scores for fracture detection, averaged to generate a collective prediction. Thresholding on these scores further refines image classification, improving accuracy and robustness in fracture identification.

The proposed ensemble model, which is underpinned by the sophisticated concept of a stacking algorithm, represents a powerful approach to predictive modeling that can substantially increase the accuracy and robustness of classification tasks. In general, ensembles are hailed as the go-to strategy when seeking to extract the best results from multiple individual models. We aimed to build a more complete and precise forecasting system by merging various models, each with its own specific techniques, biases, and feature representations.

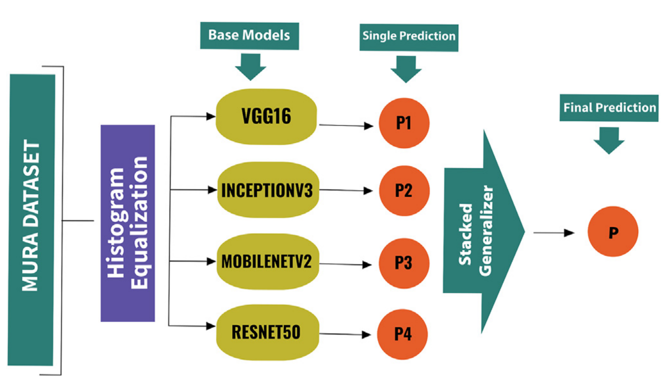

Figure 1 The proposed ensemble model (Stacking algorithm).

Each base model in our ensemble is tailored and modified to extract specific patterns and cues from these images. They delved deeply into pixel-level details, striving to identify the telltale signs of fractures. The output from each base model is not a binary classification but rather a probability score. These scores, which range from 0 to 1, express the degree of confidence of the model in assigning an input image to a positive class (indicating a fracture). Flow diagram of the proposed model is given in Fig 2.

Modified Vgg16

VGG16, developed by the Visual Geometry Group at Oxford, is a renowned CNN known for its implicity and effectiveness. It achieved top performance in the 2014 ImageNet Challenge with 16 layers, including 13 convolutional layers and 3 fully connected layers, using ReLU activation and max-pooling. The VGG16 architecture, which is renowned for its process in the realm of DL, played a pivotal role. The model is imported from the Keras library, bearing the weighty knowledge of its pretraining on a vast and diverse ImageNet dataset. Figure 3 shows the modified VGG16.

The input images were standardized to 224 x 224 pixels and fed into a VGG16 model with 14 million parameters, fine-tuning 15,000. Enhancements included Global Average

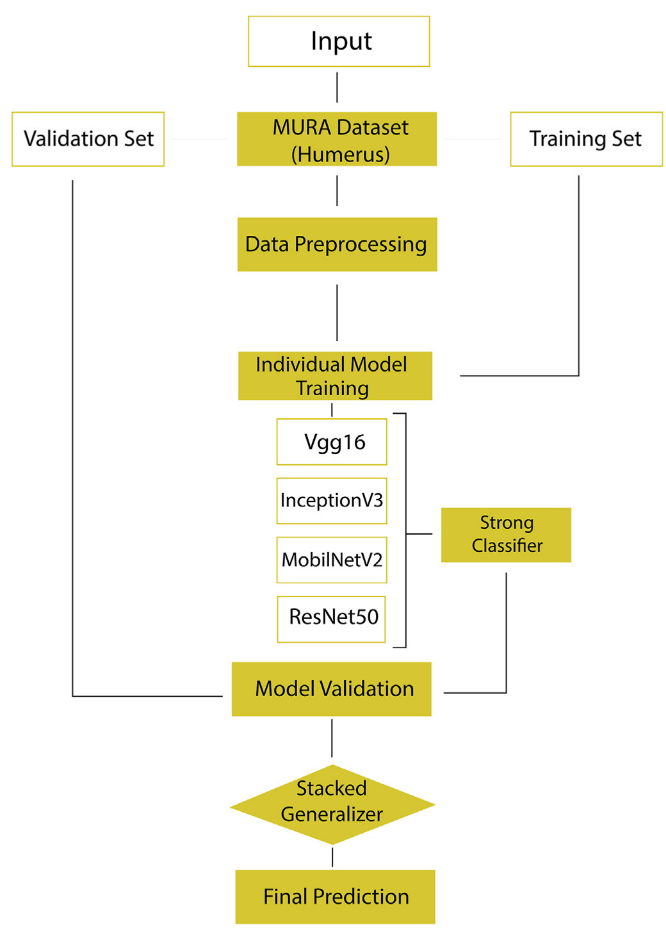

Figure 2 Flow diagram of the proposed model.

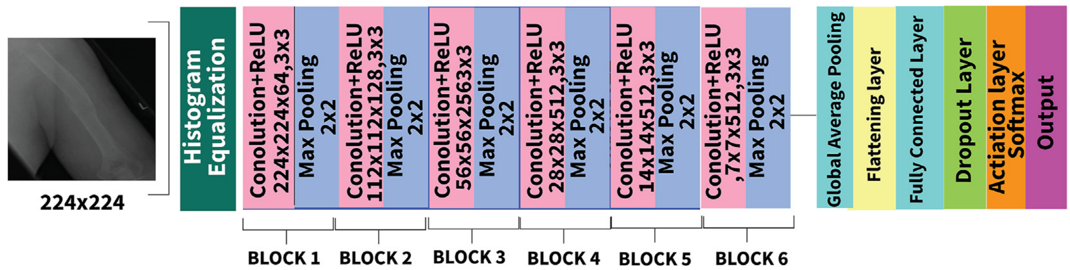

Figure 3 Modified VGG16 model architecture.

Pooling for spatial feature extraction, Flatten for preparing feature maps, and Dropout to mitigate overfitting. The model used softmax activation for binary classification, optimized over eight epochs with the Adam optimizer (learning rate 0.01, binary cross-entropy loss), resulting in a robust classifier.

Modified inceptionV3

Inception V3, developed by Google, is a highly efficient and powerful CNN renowned for its multiscale feature capture and top performance in image classification tasks such as the ImageNet Challenge. The InceptionV3 model, known for being a CNN design used for image classification was prominently featured. To establish the groundwork for a learning model, key libraries such as Keras and TensorFlow were brought in. The diagram, in Fig 4, displays the structure of the Adapted InceptionV3 Model.

Several important improvements were skillfully incorporated into the structure to boost its effectiveness. These upgrades involved the inclusion of a Global Average Pooling layer, for extracting features and a dropout layer to avoid overfitting by deactivating neurons during training. A tailored output layer was designed to cater to the requirements of the classification task, thereby elevating the models performance and adaptability. InceptionV3 was used in the ensemble model due to its sophisticated inception modules, which effectively capture multiscale features by using parallel convolutional filters of different sizes.

Modified MobilenetV2

MobileNetV2, created by Google, enhances the efficiency of MobileNetV1 for use in mobile and embedded vision applications. It introduces inverted residuals with linear bottlenecks and depthwise separable convolutions to enhance performance and reduce computational cost. The utilization of a pretrained MobileNetV2 architecture, initially primed through extensive training on the vast ImageNet dataset, took the center stage. The foundational library Keras and Tensorflow was imported, laying the essential groundwork for this DL endeavor. Figure 5 shows the architecture of the modified MobileNetV2 model architecture.

MobileNetV2, with 2.25 million nontrainable parameters from its ImageNet training, integrated a Global Average Pooling layer for spatial feature extraction. A flatten layer prepared feature maps for a fully connected layer in the binary classification context, supported by an output layer with two neurons and softmax activation. With 7,686 trainable parameters and a learning rate of 0.01, the model was finely tuned for efficient and precise classification performance. MobileNetV2 is used for its efficiency, ideal for mobile and embedded devices due to depthwise separable convolutions and inverted residuals that reduce computational cost while maintaining high accuracy.

Modified Resnet50

ResNet50 by Microsoft uses residual learning and batch normalization across 50 layers to address gradient

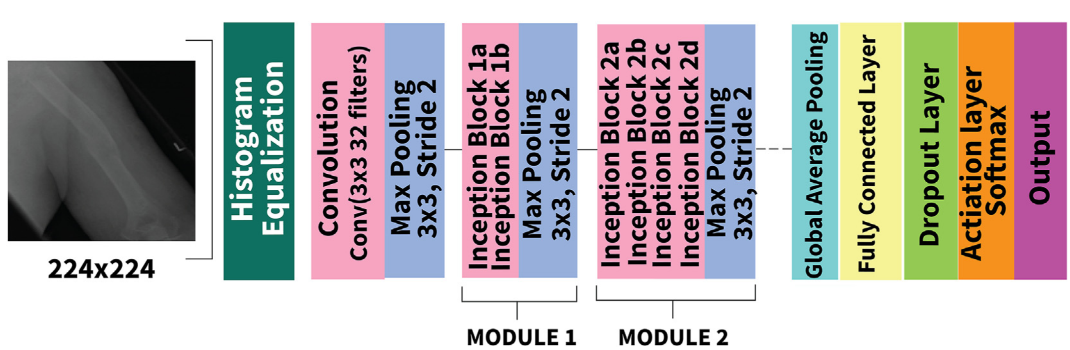

Figure 4 Modified inceptionV3 model architecture.

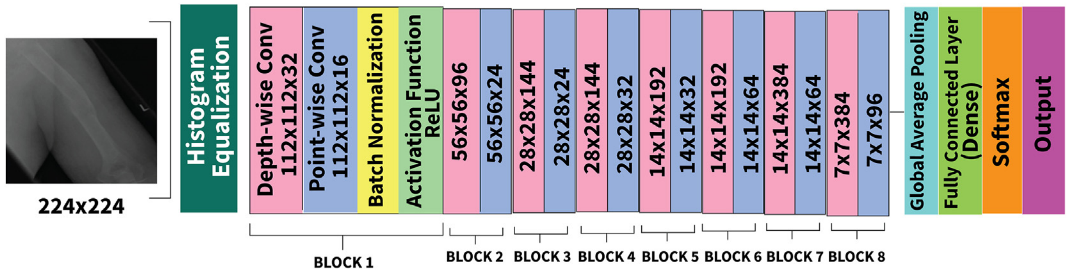

Figure 5 Modified mobileNetV2 model architecture.

vanishing, ensuring high accuracy in image classification.[33] Leveraging its robustness, our approach fine-tuned ResNet50 with 53,120 parameters while preserving 25.5 million pretrained parameters for task-specific adaptation. Figure 6 presents the modified ResNet50 model architecture.

VGG16, MobileNetV2, InceptionV3, and ResNet50 were chosen for their distinct strengths in image classification, with VGG16 and ResNet50 capturing complex patterns, MobileNetV2 offering efficiency, and InceptionV3 providing multiscale feature extraction. Integrating these models into an ensemble leverages their synergistic capabilities, enhancing overall classification performance.

Experimental setup

Data selection

In this study, we utilized the humerus images from the MURA-v1.1 dataset[34] for training and evaluating our models. The MURA dataset is a large-scale collection of musculoskeletal radiographs labeled by radiologists for fracture detection. There are a total of 6,542 radiographs of the humerus. In our study, there are two classes: fracture and nonfracture. We extracted all humerus radiographs from the MURA-v1.1 dataset. Each image in this subset was carefully selected based on the expert labels provided. This ensured that our analysis and model training would be based on high-quality, clinically validated data. Supplementary Fig 2 shows the samples from fractured class.

Histogram equalization

Histogram equalization has been applied to X-ray images to enhance contrast. Histogram equalization is an image-processing technique, improves image contrast by redistributing intensity values for a more uniform Histogram.[35] This technique enhances the visibility of subtle structures such as fractures in X-ray images, aiding radiologists in making more accurate diagnoses. Image enhancement technique, contrast enhancement, is applied to the pre-processed augmented images. Supplementary Fig 1 presents (a) original image histogram and (b) histogram of the transformed (equalized) image.

In supplementary Fig 1 (a), the histogram has values spread across a narrow range, which means there may be areas of low contrast or dynamic range in the image. In supplementary Fig 1(b), the histogram has a more uniform distribution, which means that the image's contrast has been enhanced.

Data augmentation

Data quality and quantity are crucial for DL model performance. To address limited data, the study used data augmentation techniques, including random rotations (-20 to +20 degrees), horizontal flips (with a 50% probability), and zooming (0.8–1.2 times the original size) to increase diversity and improve model generalization without collecting new samples.

Data augmentation is performed after histogram equalization to increase the dataset size, resulting in 15,978 X-ray images. The training dataset contained 12,796 images, of

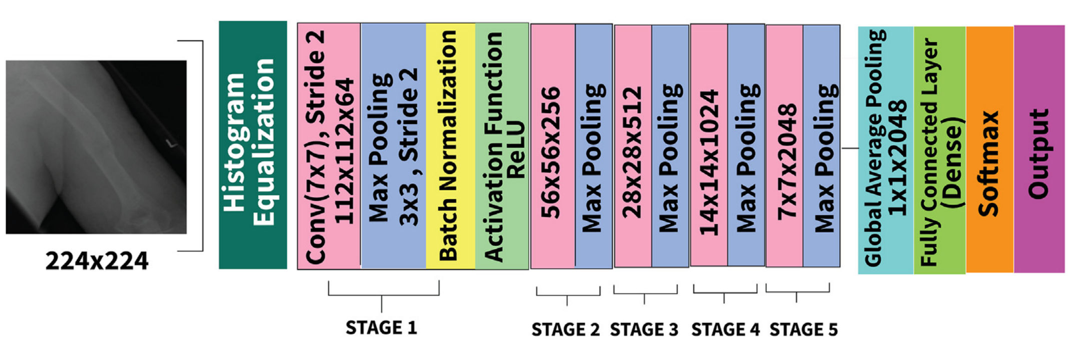

Figure 6 Modified ResNet50 model architecture.

Table 1
Training and validation dataset.

	Fractured	Non Fractured	Total
Training dataset	6,402	6,394	12,796
Validation dataset	1,576	1,606	3,182

which 6,402 were fractured and 6,394 were nonfractured. In the validation dataset, there were 3,180 images in total: 1,576 were fractured and 1,606 were nonfractured, as shown in the Table 1. Supplementary Fig 3 (a)–(c) show samples of augmented images.

Data splitting

After data augmentation, a single training–validation split is utilized. A postaugmentation split prevents augmented data from leaking into the validation set, preserving evaluation integrity. A single training–validation split divides the dataset into 70–80% for training and 20–30% for validation, optimizing parameters and assessing performance on unseen data. This method simplifies workflow, aids in hyperparameter tuning, and ensures robust model development before final evaluation on an independent test set. The dataset is divided at a ratio of 80:20.

Performance measures

The effectiveness of the proposed ensemble model is assessed using the F1 score, a metric that strikes a balance between precision and recall. This evaluation method proves valuable in scenarios involving classification tasks such as X-ray image categorization, where accurate predictions for both negative outcomes are important but carry different significance. By accounting for positives and false negatives, the F1 score offers an evaluation of the models precision in detecting fractures in medical images.

1) Equation (1) states that precision is defined as true positive divided by the sum of true positive and false positive.
2) According to equation (2), recall is calculated by dividing true positive by the total of true positives and false negatives.
3) The F1 score is calculated using two precision times. Recall is subtracted from the product of precision and recall, as shown in equation (3):

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \tag{1}$$

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \tag{2}$$

$$F1\ Score = \frac{PrecisionXRecall}{Precision + Recall} \tag{3}$$

In this study, models were trained and evaluated using Google Colab, a cloud-based platform that provides graphics processing unit (GPU) resources to accelerate the training of learning models. We chose to utilize Colab because of its access, to these GPU resources, which greatly improved the efficiency of our model training, ultimately enhancing the reliability and scalability of our experiments. Accuracy played a role in assessing how well our classification models performed as it gives an indication of predictive accuracy. In our study, accuracy was calculated as follows:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \tag{4}$$

During the validation phase, we computed the accuracy, for each of the models VGG16, MobileNetV2, InceptionV3, and ResNet50. Accuracy was selected for its simplicity and effectiveness in evaluating model performance, reflecting the proportion of correct predictions made by the model, as described in equation 4. Additionally, accuracy served as a basis for comparing the models' performance under similar training conditions.

Results and discussion

VGG16, InceptionV3, MobileNetV2, and ResNet50 were trained using Keras, with pretraining on ImageNet. VGG16 had 15,000 trainable parameters, MobileNetV2 had 7,686, and ResNet50 had 53,120, all fine-tuned with a learning rate of 0.01 and trained over 8 epochs to prevent overfitting. Using the same number of epochs for all models ensured consistent training for fair performance comparison. The proposed ensemble model achieved 94.02% training and 92.96% validation accuracies, whereas it achieved a training loss of 0.243 and a validating loss of 0.224. All the comparisons graphs of the proposed methodology are given as supplementary Fig 4 to Supplementary Fig 9. VGG16 achieved a training accuracy of 82.5% and a validation accuracy of 82.2% after 8 epochs, indicating good but slightly overfitted performance. InceptionV3 showed a training accuracy of 83.6% and a validation accuracy of 81%, demonstrating strong generalization despite a minor drop in validation accuracy. MobileNetV2 converged well with a training and validation loss of 0.2, achieving 90% training accuracy and 88% validation accuracy, highlighting effective learning and generalization capabilities. ResNet50 exhibited stable convergence with a training accuracy of 86% and a validation accuracy of 86% after 8 epochs, showcasing robust performance in learning intricate patterns within the training data.

Comparison histogram of accuracy, precision, recall, and F1-Score of all the existing models used in the proposed method for the classification of bone abnormalities is shown in Fig 7. Comparison results of the proposed method with the existing state-of-the-art techniques are given in Table 2.

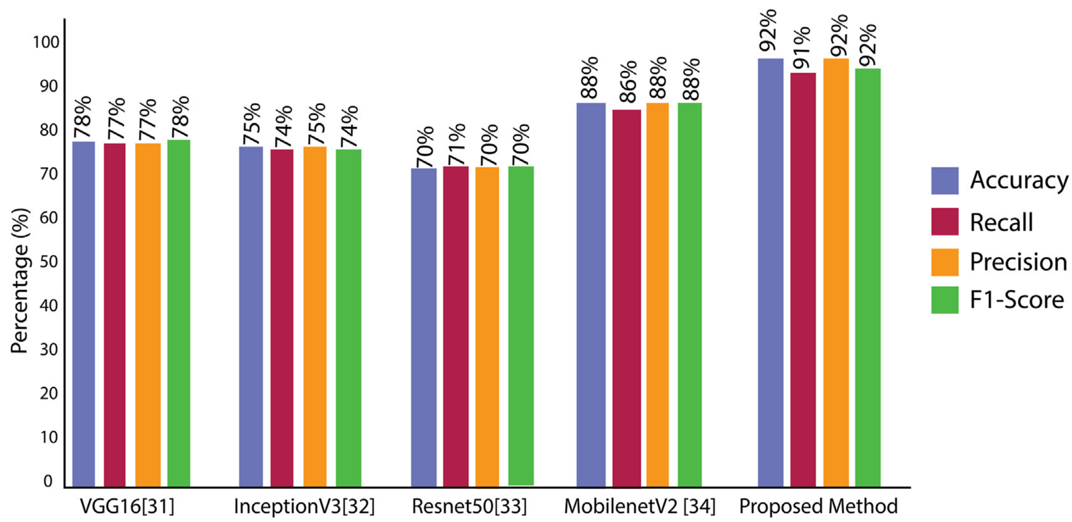

Figure 7 Comparison histogram of existing techniques with proposed technique.

Table 2
Comparison analysis of existing techniques with proposed model.

Methods	Accuracy	Precision	Recall	F1-Score
DenseNet-201[32]	0.88	0.84	0.84	0.88
DenseNet-169[32]	0.84	0.85	0.84	0.83
InceptionResNetV2[32]	0.86	0.83	0.84	0.87
VGG16[36]	0.78	0.77	0.77	0.78
InceptionV3[33]	0.75	0.74	0.75	0.74
ResNet[37]	0.70	0.71	0.70	0.70
MobileNetV2[38]	0.88	0.86	0.88	0.88
ModifiedPro posed VGG16	0.82	0.83	0.83	0.82
Modified Proposed InceptionV3	0.81	0.81	0.81	0.81
Modified Proposed ResNet	0.90	0.90	0.89	0.90
Modified Proposed InceptionV3	0.81	0.81	0.81	0.81
Modified Proposed ResNet	0.90	0.90	0.89	0.90
Modified Proposed MobileNetV2	0.74	0.73	0.74	0.75
Proposed Ensemble Model	0.92	0.91	0.91	0.92

The proposed ensemble model, despite its robust architecture and promising performance metrics, has several limitations including increased computational complexity and training time due to integrating multiple DL architectures, extensive experimentation required for fine-tuning hyperparameters, and challenges in scaling to larger datasets or different domains. Potential biases can emerge from imbalances in the training data; inherent biases in pre-trained models such as VGG16, InceptionV3, MobileNetV2, and ResNet50 are two models that have been considered, along with the choices made during feature selection and preprocessing. There is a concern about how the model can adapt to datasets as its performance might decrease, requiring substantial modifications for various tasks or real world uses. Moreover, statistical uncertainty arises from differences in training validation splits and the random nature of the training process making metrics, such as accuracy and F1 score, not entirely indicative of error distribution.

Conclusion

The proposed model in this study showcases an efficient method, for improving accuracy and adaptability in tasks related to DL. By leveraging the advantages of models minimizing overfitting and enhancing reliability, it emerges as a choice for real-world use. The flexibility and adjustability of the model also highlight its promise as an instrument for addressing challenges in different fields.

Ethics

Not required because the dataset used is a public dataset.

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Author contribution

1. Guarantor of integrity of the entire study: Ayesha Tahir
2. Study concepts and design: Ayesha Tahir, Ayesha Saadia, khurram khan
3. Literature research: Ayesha Tahir, Ammara Gul, Ayesha Saadia
4. Clinical studies: Raja Naeem Akram, Ayesha Saadia
5. Experimental studies/data analysis: Ayesha Tahir, Ayman
6. Statistical analysis: Ayesha Tahir, Ayesha Saadia, khrram khan

7. Manuscript preparation: Ayesha Tahir, Ayesha SAadia, Ammara Gul, khurram khan
8. Manuscript editing: Ayesha Saadia, Ayman

Conflict of interest

The authors declare no conflict of interest.

Data availability

The authors declared that the datasets used in this research are publicly available.

Institutional review board statement

Not applicable.

Informed consent statement

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.crad.2024.08.006.

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